

# Gauss-Jordan Elimination

Ex 1 Solve  $x - y + 2z - w = -1$

by Gauss-Jordan elimination.

$$2x + y - 2z - 2w = -2$$

$$-x + 2y - 4z + w = 1$$

$$3x - 3w = -3$$

Sol The augmented matrix is

$$\left[ \begin{array}{cccc|c} 1 & -1 & 2 & -1 & -1 \\ 2 & 1 & -2 & -2 & -2 \\ -1 & 2 & -4 & 1 & 1 \\ 3 & 0 & 0 & -3 & -3 \end{array} \right]$$

$$\begin{array}{l} r_2' = 2r_1 - r_2 \\ r_3' = r_1 + r_3 \\ r_4' = 3r_1 - r_4 \end{array} \left[ \begin{array}{cccc|c} 1 & -1 & 2 & -1 & -1 \\ 0 & -3 & 6 & 0 & 0 \\ 0 & 1 & -2 & 0 & 0 \\ 0 & -3 & 6 & 0 & 0 \end{array} \right]$$

$$\underline{r_2' = r_2 / -3} \rightarrow \left[ \begin{array}{cccc|c} 1 & -1 & 2 & -1 & -1 \\ 0 & 1 & -2 & 0 & 0 \\ 0 & 1 & -2 & 0 & 0 \\ 0 & -3 & 6 & 0 & 0 \end{array} \right]$$

$$\begin{array}{l} r'_1 = r_1 + r_2 \\ r'_3 = r_2 - r_3 \\ r'_4 = 3r_2 + r_4 \end{array} \quad \left[ \begin{array}{cccc|c} 1 & 0 & 0 & -1 & -1 \\ 0 & 1 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

The above procedure produces a reduced echelon form.

The corresponding system of equations

$$x - w = -1 \quad \text{--- (i)}$$

$$y - 2z = 0 \quad \text{--- (ii)}$$

$$\text{Let } w = t, \quad z = s$$

$$\begin{aligned} \text{(ii)} \Rightarrow y - 2s &= 0 \\ \Rightarrow y &= 2s \end{aligned}$$

$$\begin{aligned} \text{(i)} \Rightarrow x - t &= -1 \\ \Rightarrow x &= t - 1 \end{aligned}$$

The general solution is

$$x = t - 1, \quad y = 2s, \quad z = s, \quad w = t$$

(6)

$$\begin{aligned} 2. \quad & -2b + 3c = 1 \\ & 3a + 6b - 3c = -2 \\ & 6a + 6b + 3c = 5 \end{aligned}$$

Sol. The augmented matrix is

$$\left[ \begin{array}{ccc|c} 0 & -2 & 3 & 1 \\ 3 & 6 & -3 & -2 \\ 6 & 6 & 3 & 5 \end{array} \right]$$

interchanging  $r_1$  and  $r_2$   $\rightarrow$

$$\left[ \begin{array}{ccc|c} 3 & 6 & -3 & -2 \\ 0 & -2 & 3 & 1 \\ 6 & 6 & 3 & 5 \end{array} \right]$$

$r'_1 = r_1/3 \rightarrow$

$$\left[ \begin{array}{ccc|c} 1 & 2 & -1 & -2/3 \\ 0 & -2 & 3 & 1 \\ 6 & 6 & 3 & 5 \end{array} \right]$$

$r'_3 = 6r_1 - r_3 \rightarrow$

$$\left[ \begin{array}{ccc|c} 1 & 2 & -1 & -2/3 \\ 0 & -2 & 3 & 1 \\ 0 & 6 & -9 & -9 \end{array} \right]$$

$$\underline{r_2' = r_2 / -2} \rightarrow \begin{bmatrix} 1 & 2 & -1 & | & -2/3 \\ 0 & 1 & -3/2 & | & -1/2 \\ 0 & 6 & -9 & | & -9 \end{bmatrix}$$

$$\begin{array}{l} \underline{r_1' = r_1 - 2r_2} \\ \underline{r_3' = 6r_2 - r_3} \end{array} \begin{bmatrix} 1 & 0 & 2 & | & 1/3 \\ 0 & 1 & -3/2 & | & -1/2 \\ 0 & 0 & 0 & | & 6 \end{bmatrix}$$

The above procedure produces a reduced row-echelon form.

The corresponding system of equations

$$a + 2c = 1/3$$

$$b - 3/2c = -1/2$$

$$0 = 6, \text{ It is not true}$$

∴ There is no solution

∴ The system is inconsistent.